Deep Learning & Applied AI

Data, features, and embeddings

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SAPIENZA
UNIVERSITÀ DI ROMA

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Data awareness

Machine learning involves dealing with data.

What do you do when you have a problem involving data?

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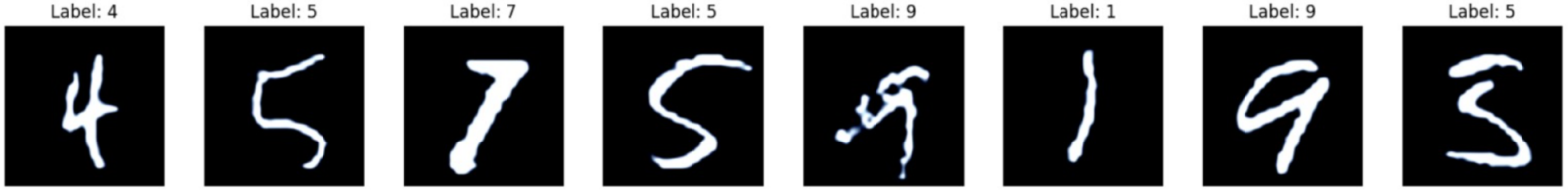
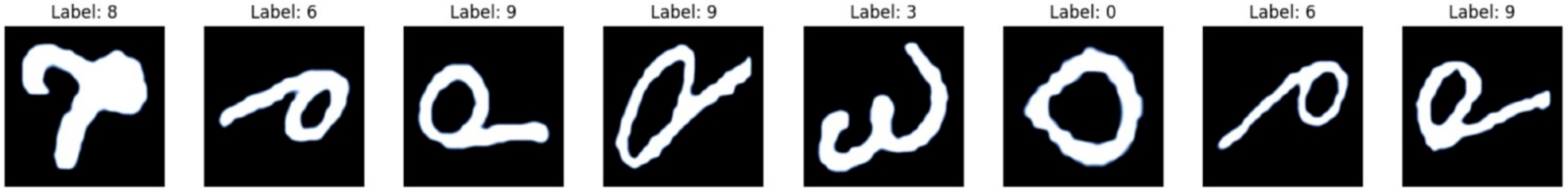
First thing: **look at the data!**

Example: Rotated data

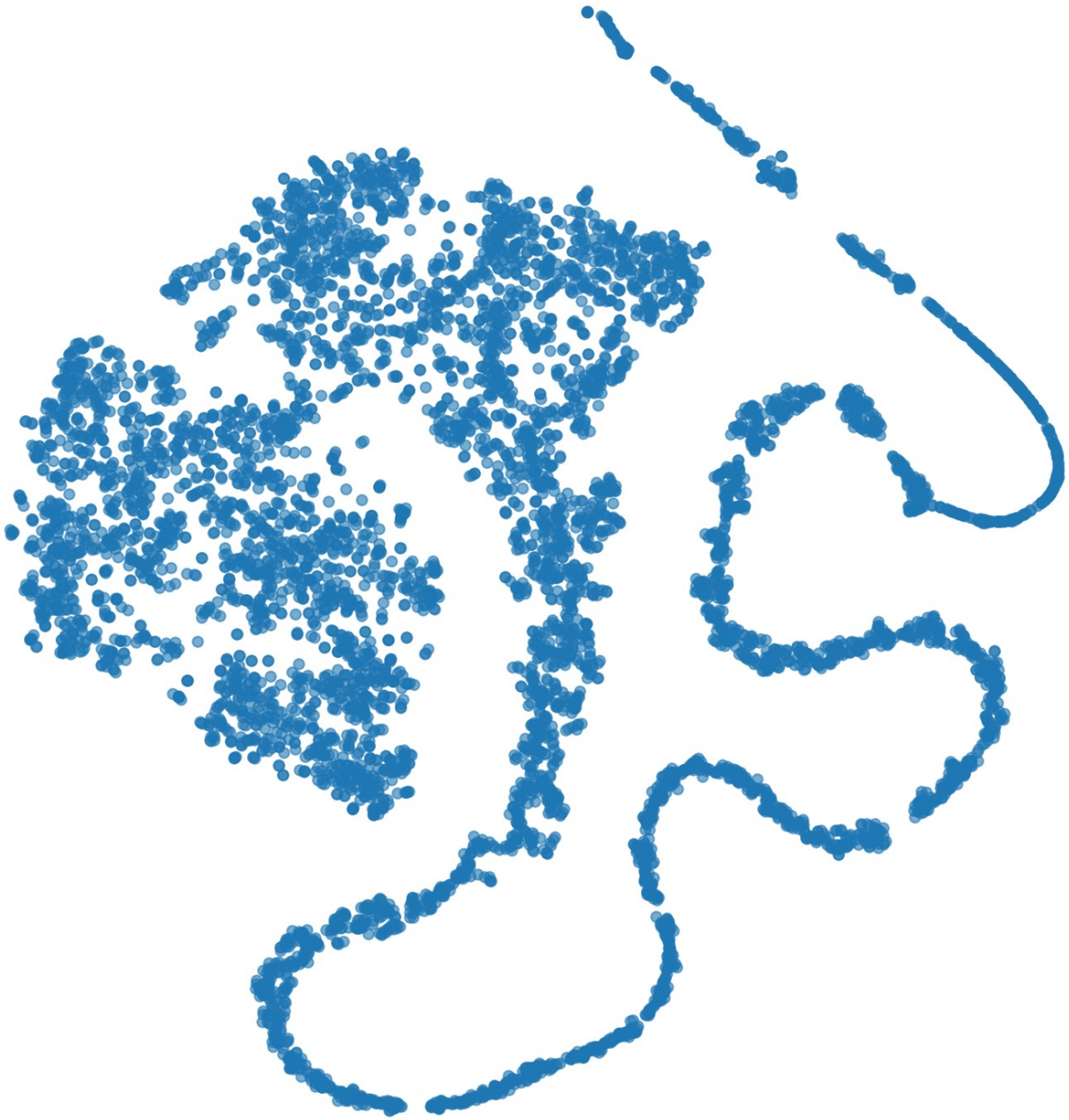
MNIST



EMNIST

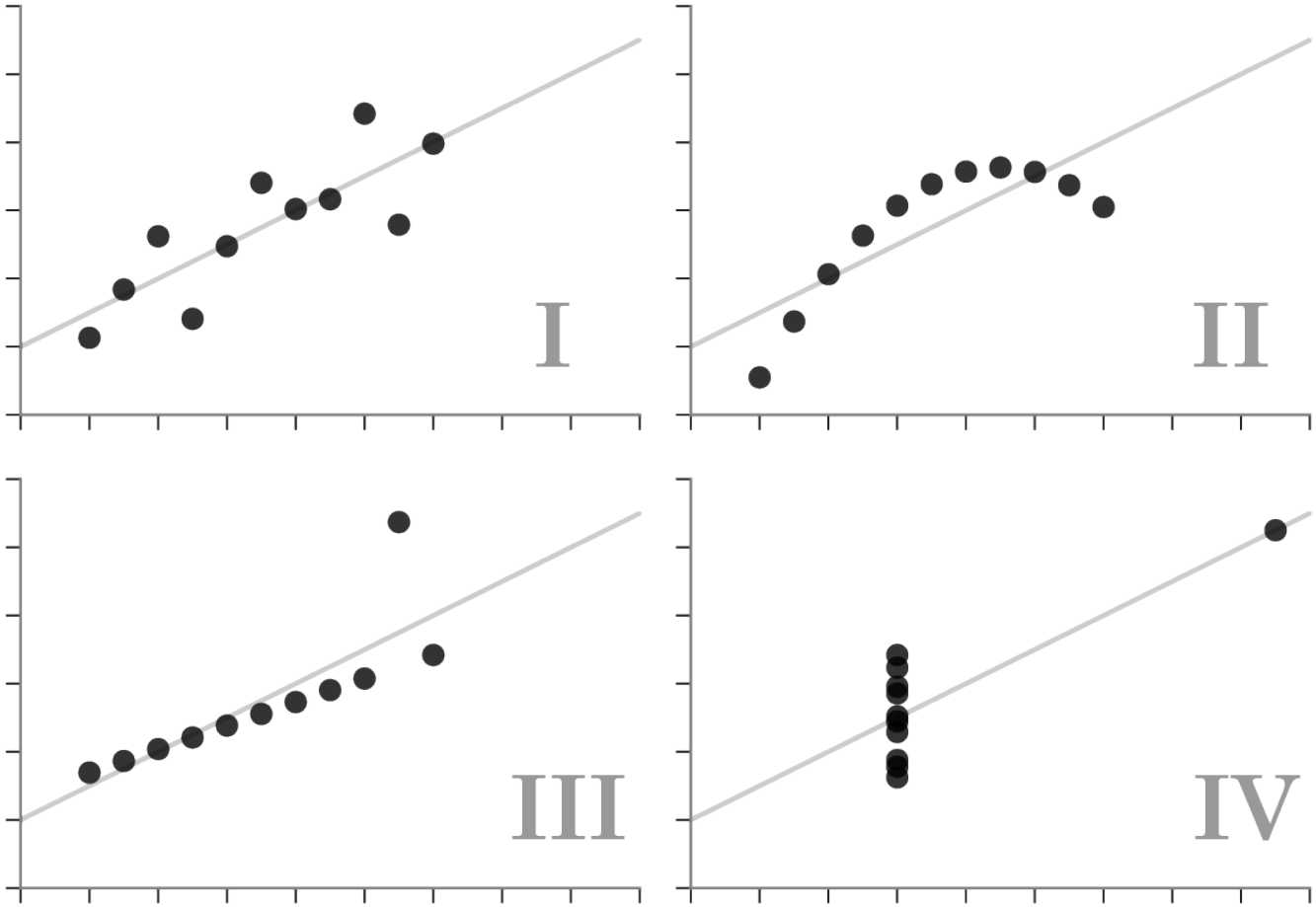


Example: Sparse dataset



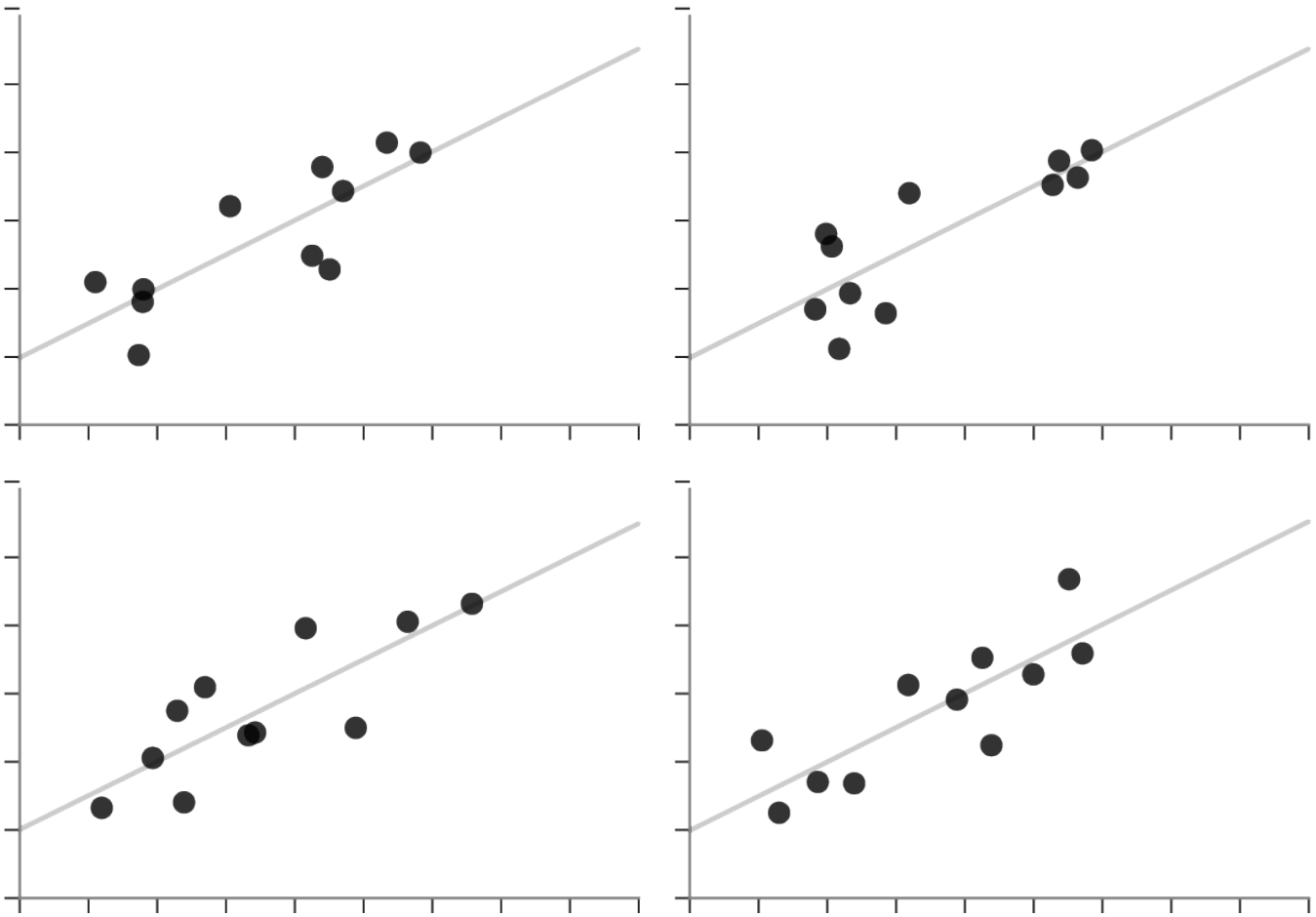
Anscombe's Quartet

Each dataset has the same summary statistics (mean, standard deviation, correlation), and the datasets are *clearly different*, and *visually distinct*.



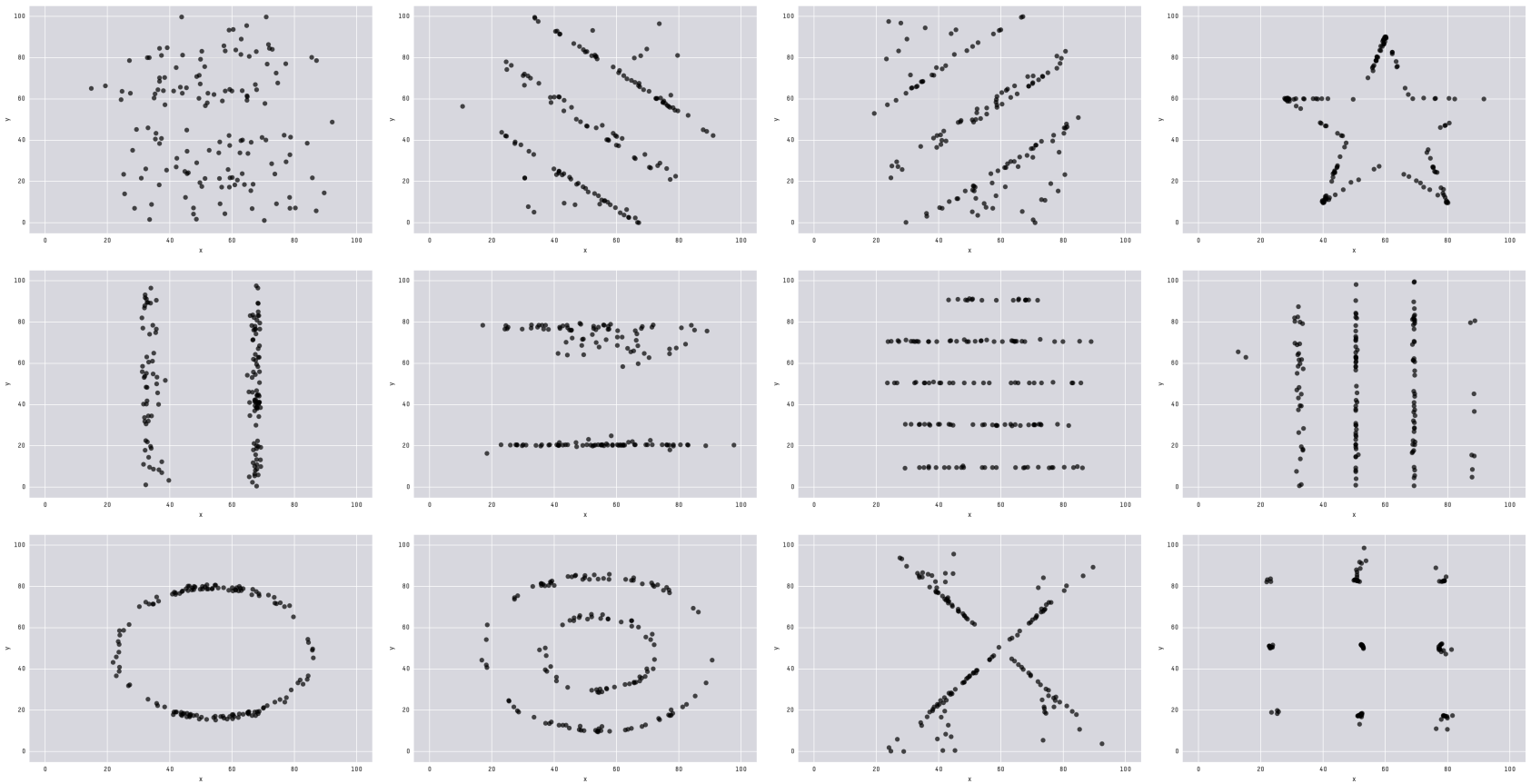
Unstructured Quartet

Each dataset here also has the same summary statistics. However, they are not *clearly different* or *visually distinct*.



The datasaurus dozen

All these datasets have the same summary stats to 2 decimal places:



Matejka and Fitzmaurice, SIGCHI 2017

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It will not always be easy to visualize.

Difficult cases: high-dimensional data, no physical access to data, implicit access to data (e.g. latent spaces).

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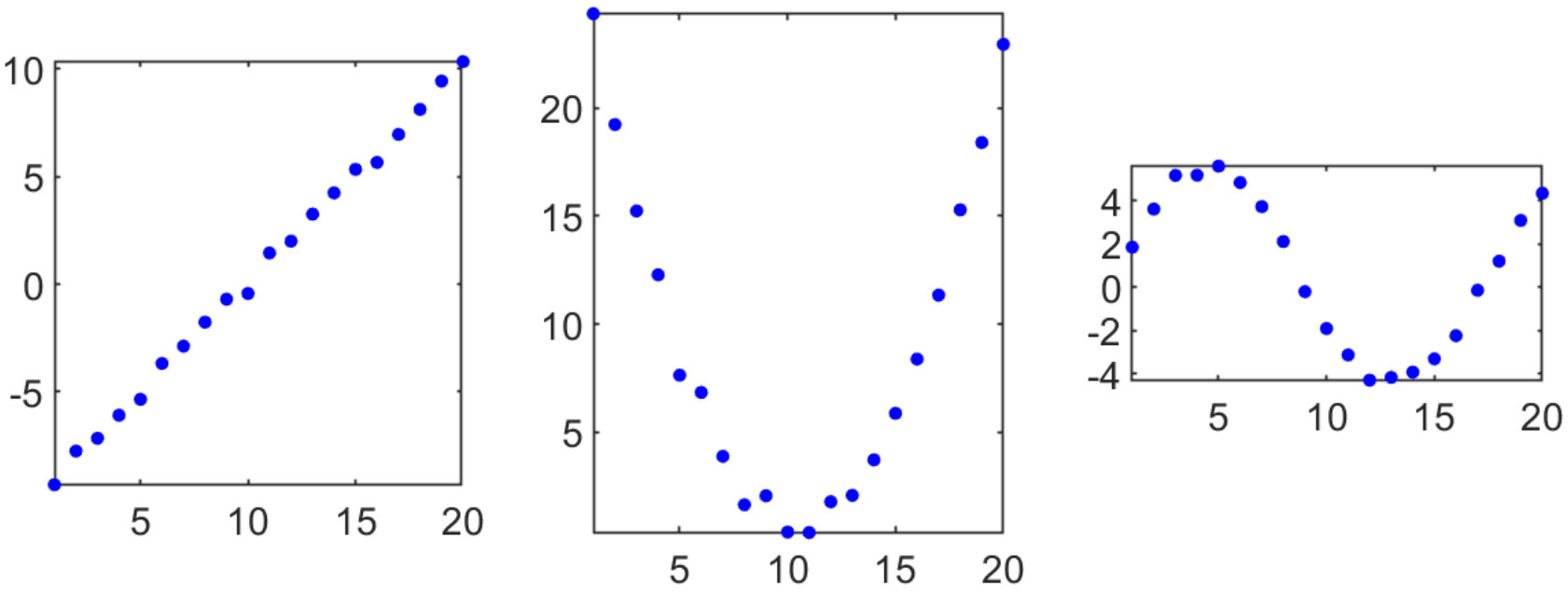

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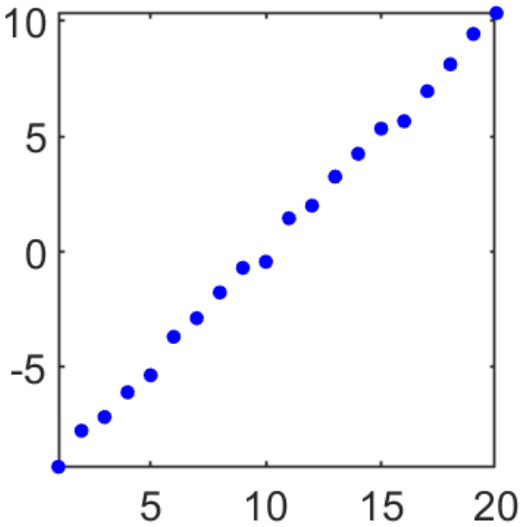
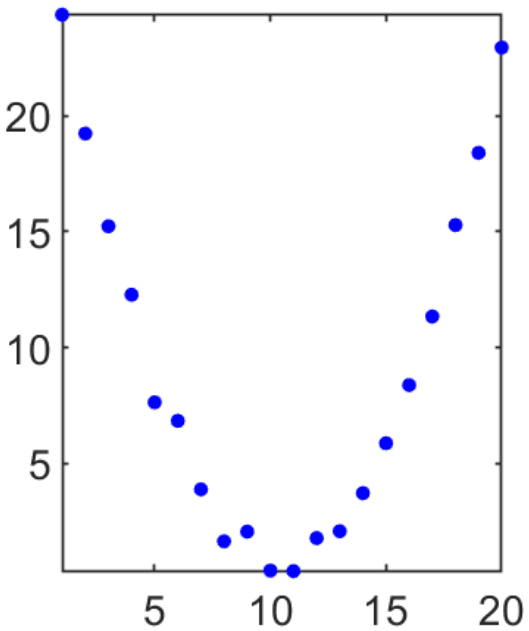


$y = ax + b$



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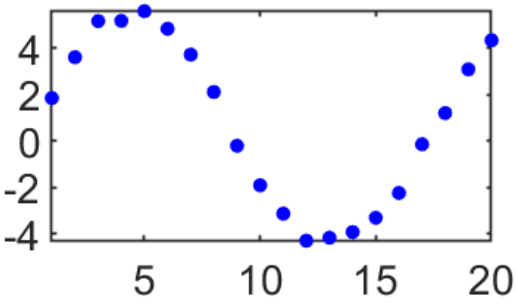

$y = ax^3 + bx^2 + cx + d$

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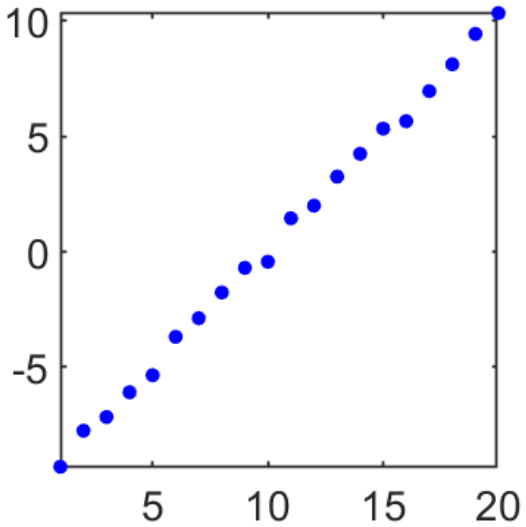

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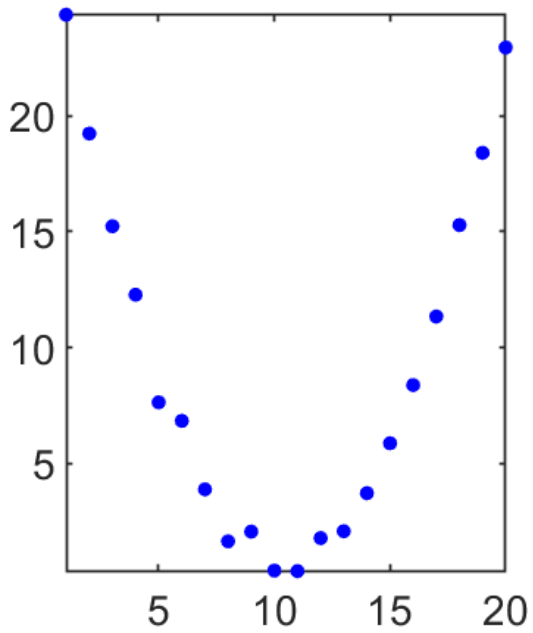

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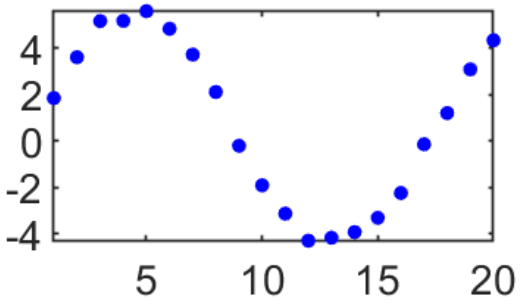

$y = a\,\mathtt{sin}(x) + bx + c$

Our model might use prior knowledge on the data.

Third plot: the data comes from a periodic process.

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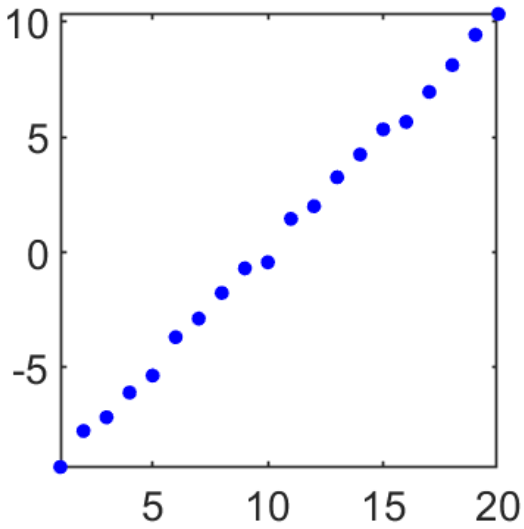

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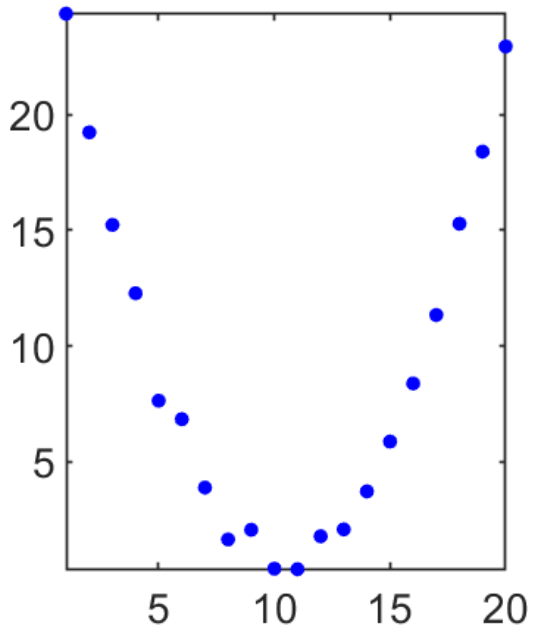

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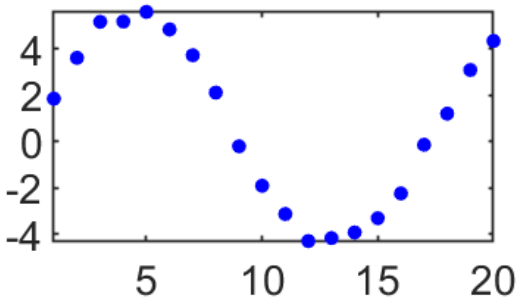

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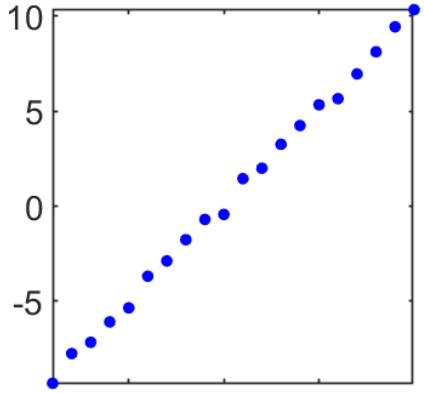

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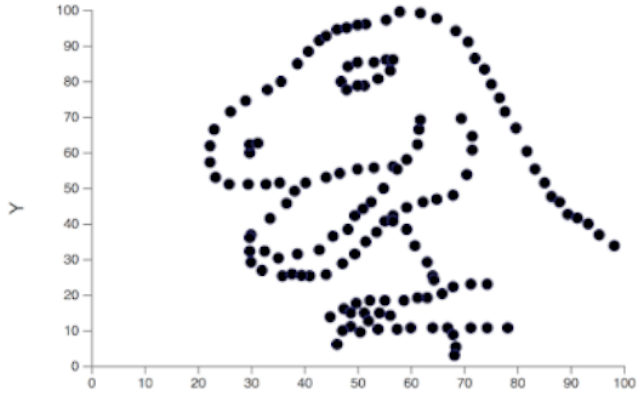

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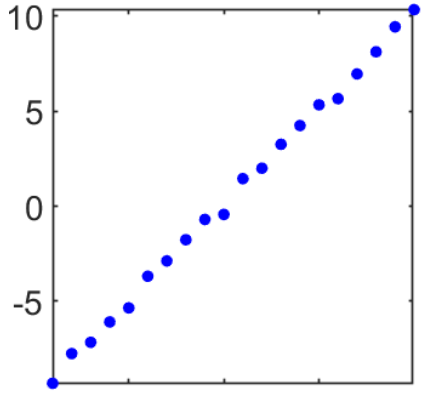
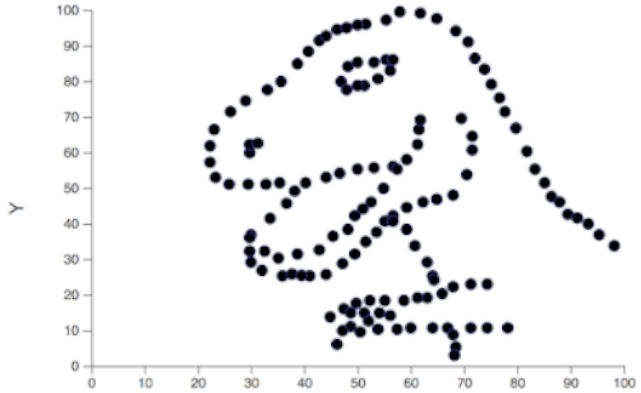


$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_x \\ a_y \end{pmatrix} t + \begin{pmatrix} b_x \\ b_y \end{pmatrix}$$



(not a function in 1D)

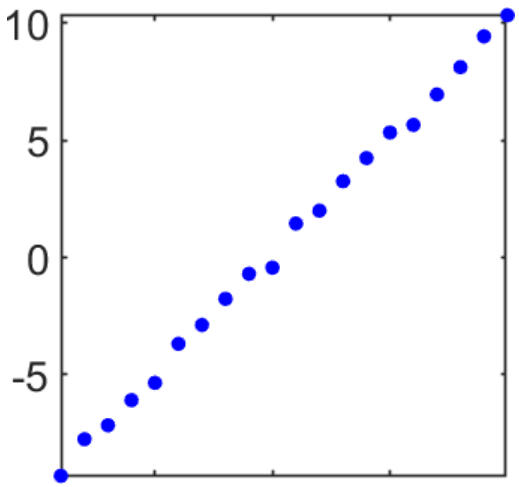
Clearly, data is not always one-dimensional.

Choosing a representation

The same data can be described in different ways.

What is the "right" way?

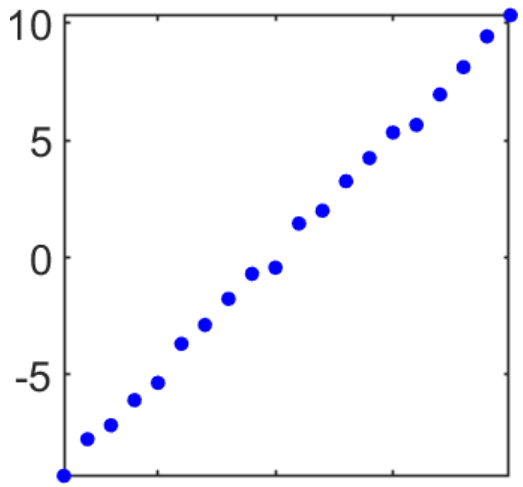
as a **function**



$$y = ax + b$$

$=$

as a **parametric curve**



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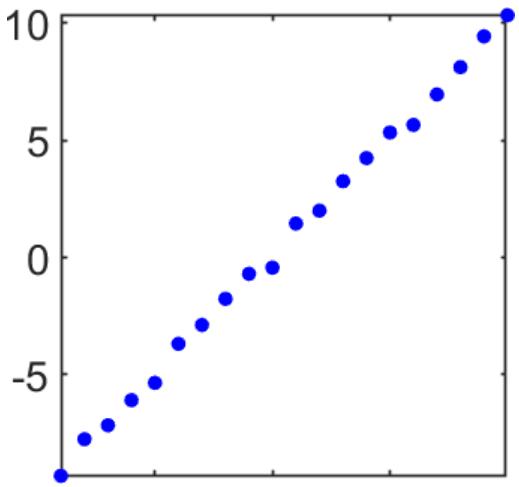
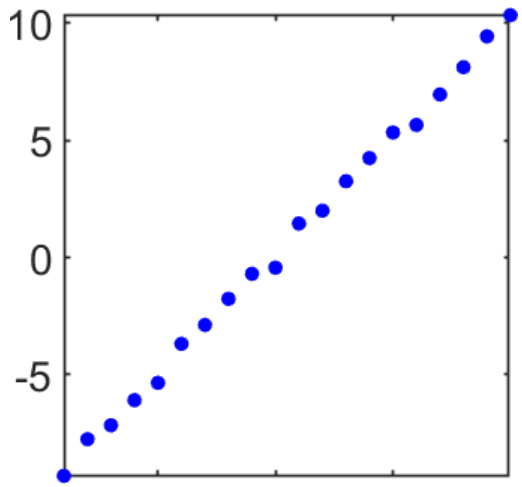


$$y = ax + b$$

2 weights

$=$

as a **parametric curve**



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4 weights

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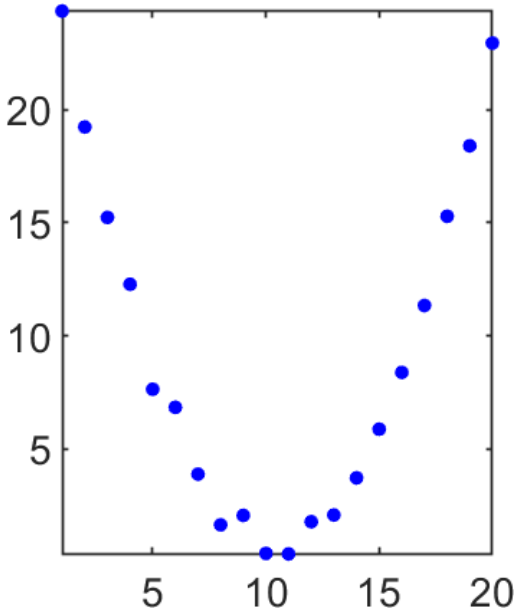

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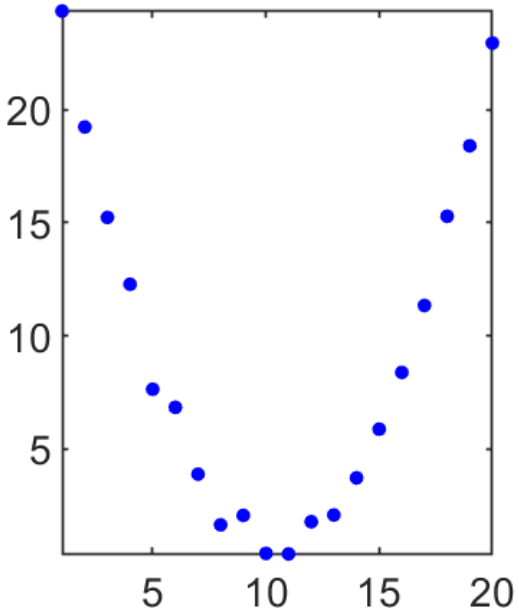

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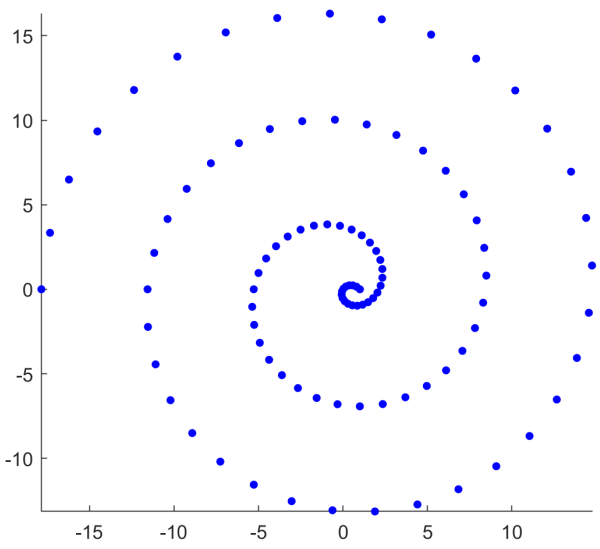

=

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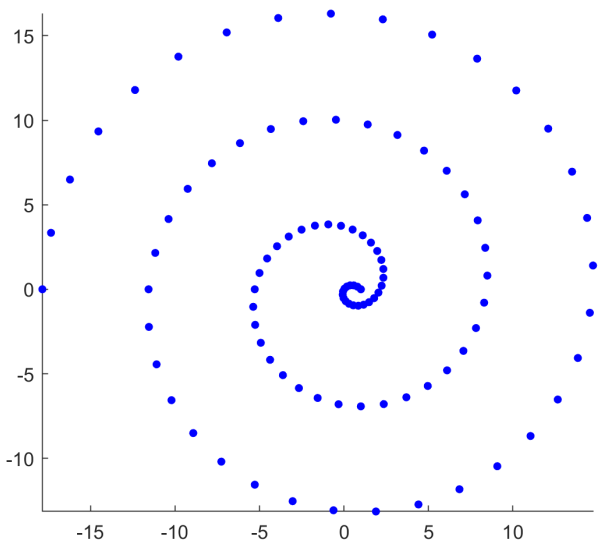

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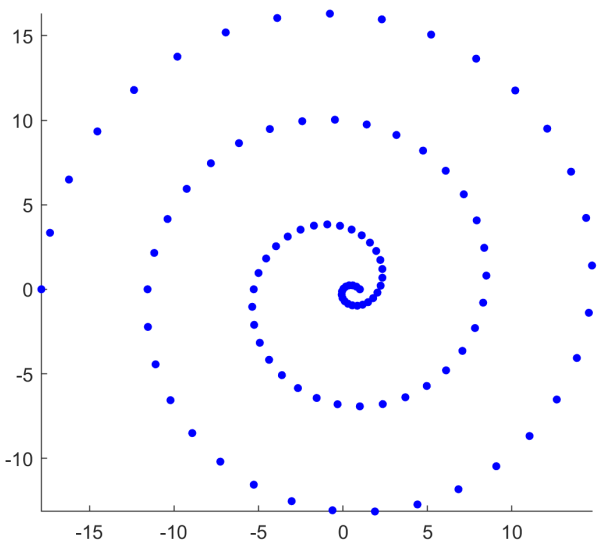

y is not a function of x

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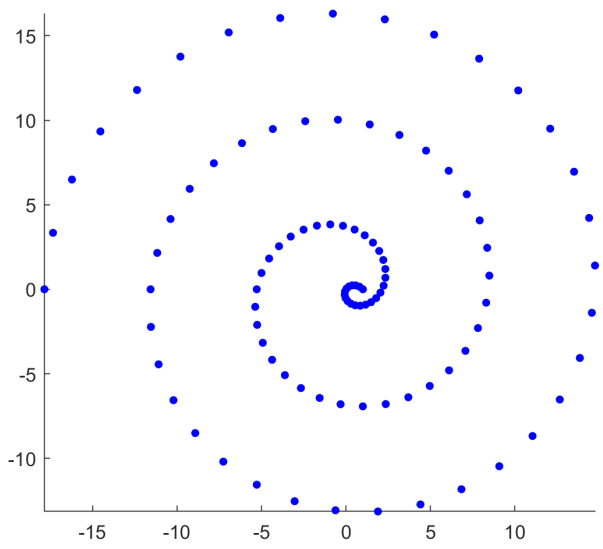

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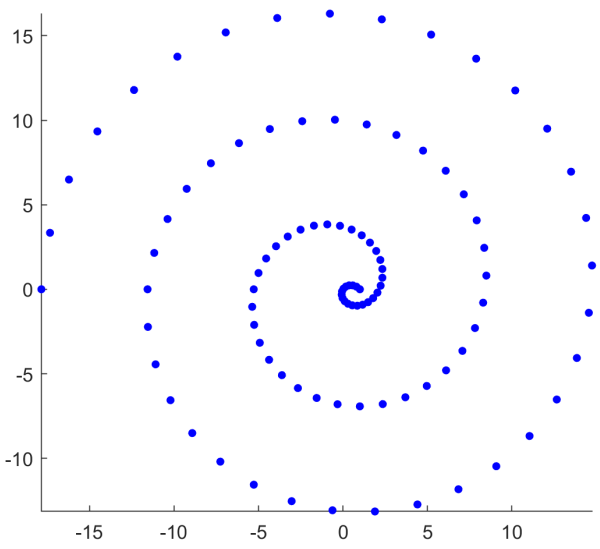

$$r = a\theta$$

(polar coordinates)

$=$

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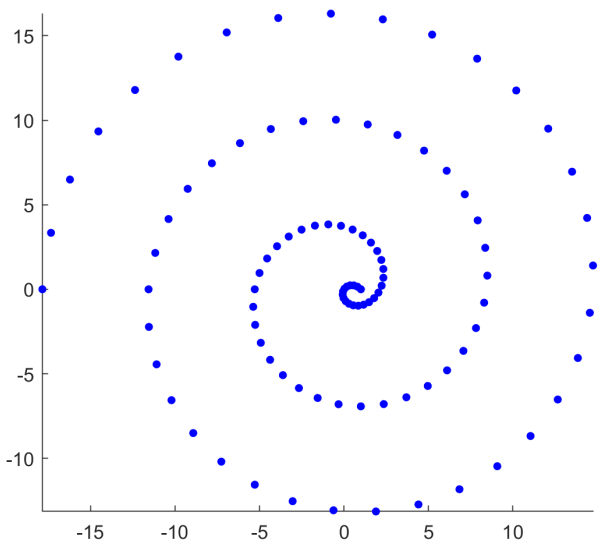

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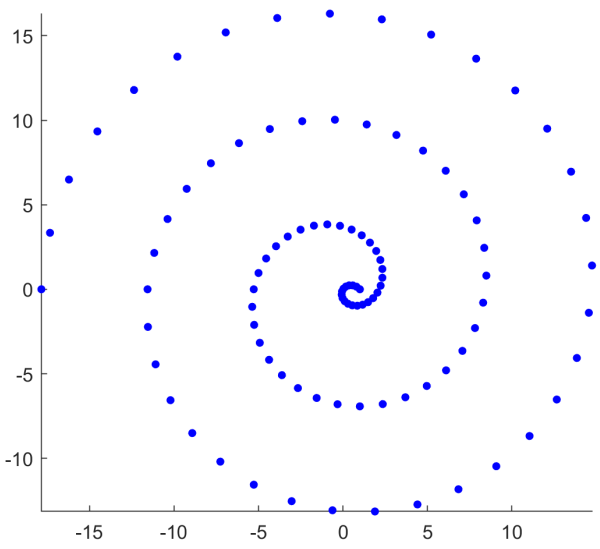

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linear!

$=$

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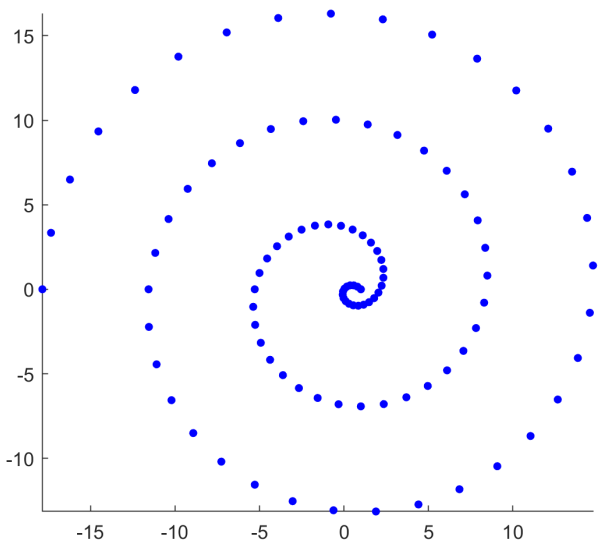

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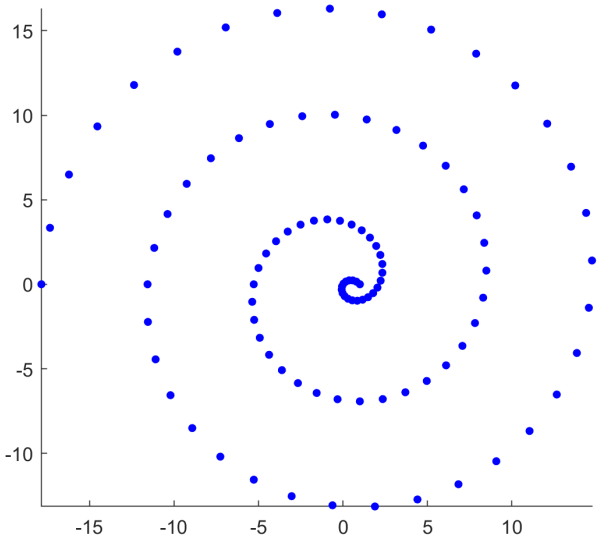

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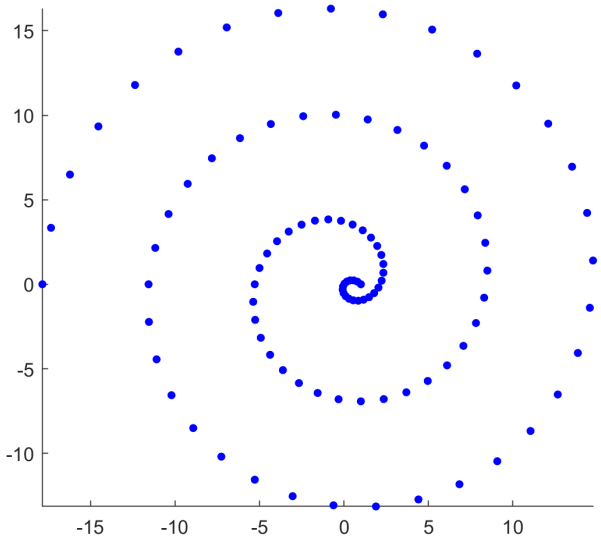

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Trade-off between #weights and simplicity

The curse of dimensionality

Of course, data can have more than 1 or 2 dimensions.

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For example, a $w \times h$ image has wh dimensions.

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Example: $\sim$1 megapixel photo (grayscale) has $\sim 10^6$ dimensions.

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The value of each coordinate is given by the gray value at that pixel. Then, the entire image is one point in a wh-dimensional space.

$\in \mathbb{R}^{w \times h} \cong \mathbb{R}^{wh}$

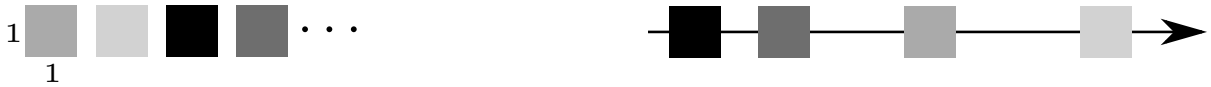
Example: $\sim$1 megapixel photo (grayscale) has $\sim 10^6$ dimensions.

Are all those dimensions significant?

The curse of dimensionality

For simplicity, consider 1×1 images, i.e., consisting of one single pixel.

Each image is a point in one dimension.



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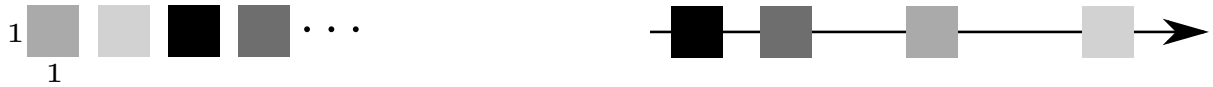
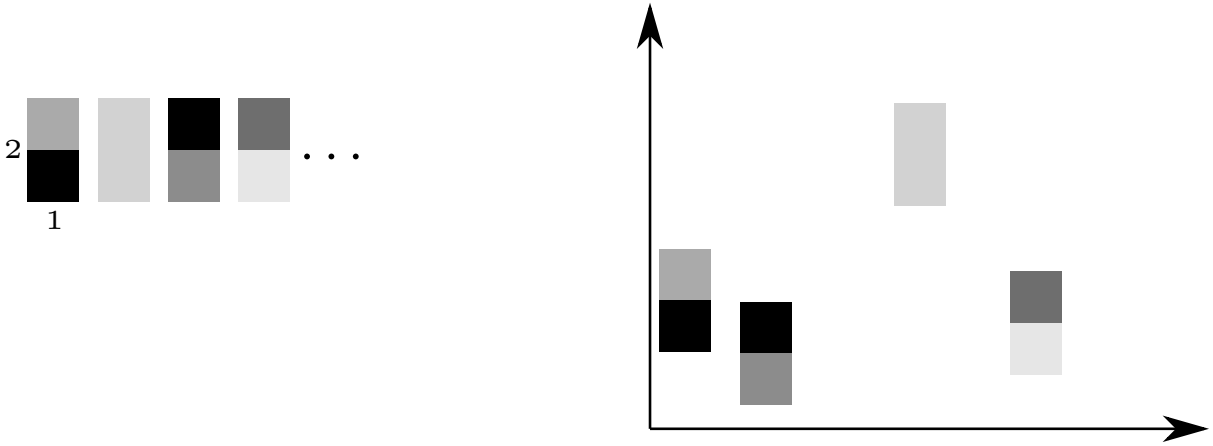


Similarly, with 2 pixels we get:



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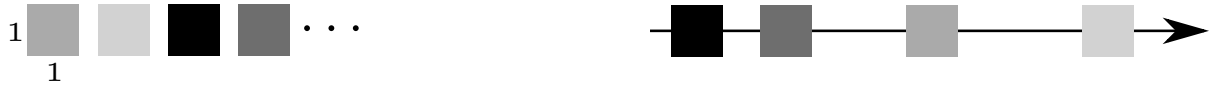
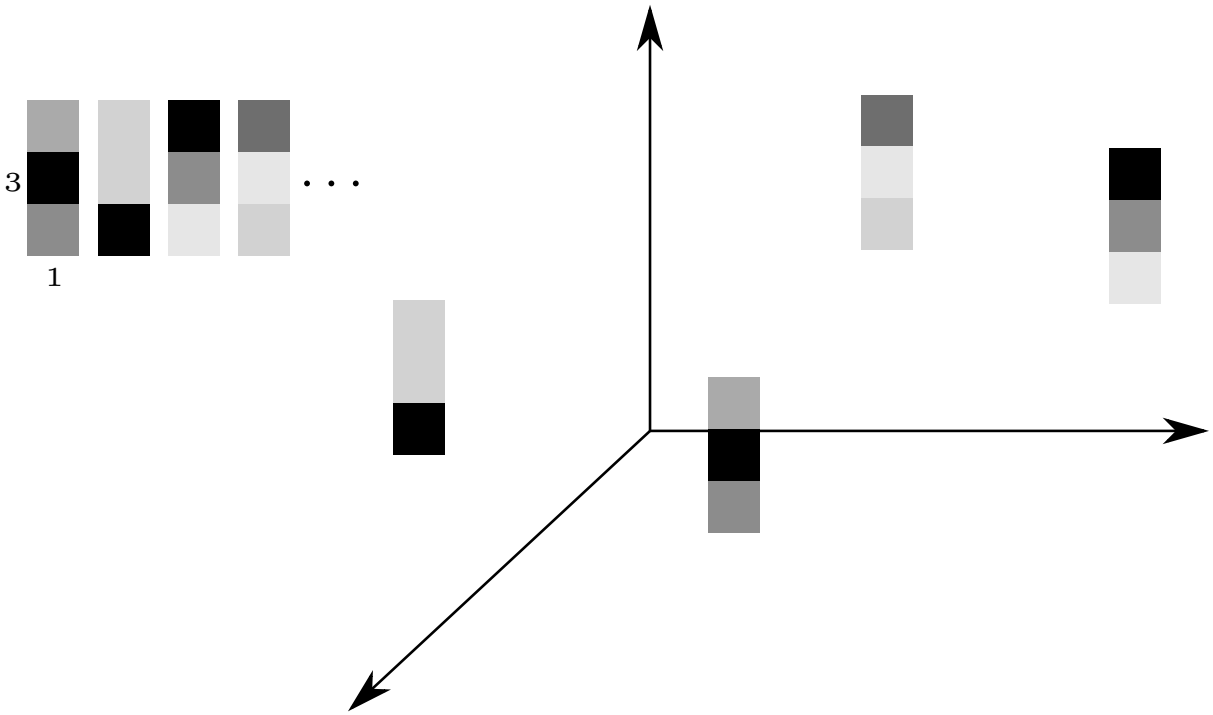


Similarly, with 3 pixels we get:



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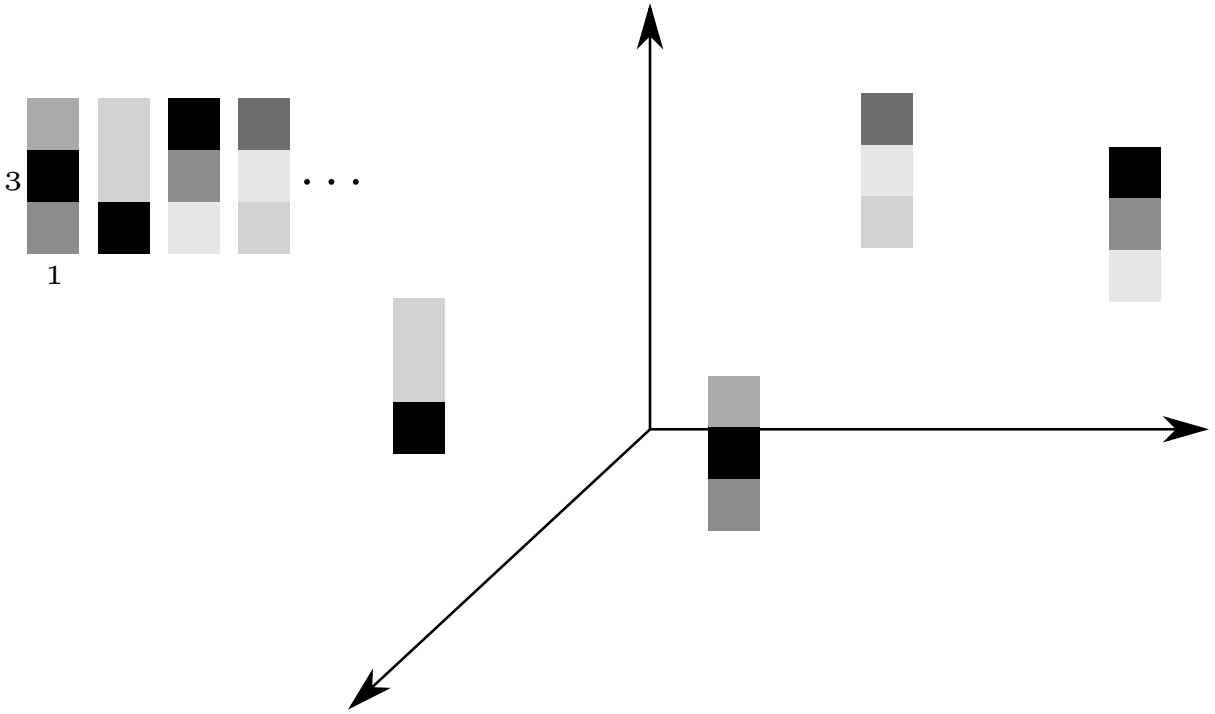
For simplicity, consider 1×1 images, i.e., consisting of one single pixel.

Each image is a point in one dimension.

Similarly, with 3 pixels we get:



Each new dimension increases sparsity of the point cloud.

The curse of dimensionality

A dataset of natural images will be extremely sparse in $\mathbb{R}^{w \times h}$, since each region of space is observed very infrequently.

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More data points make interesting structures emerge

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To cover the space entirely, we need exponentially many observations as we have dimensions!

If n data points cover well the space of 1-dim. images, then n^d data points are required for d-dim. images.

Two options:

1. **Increase** the dataset
2. **Decrease** the dimensions

The manifold hypothesis

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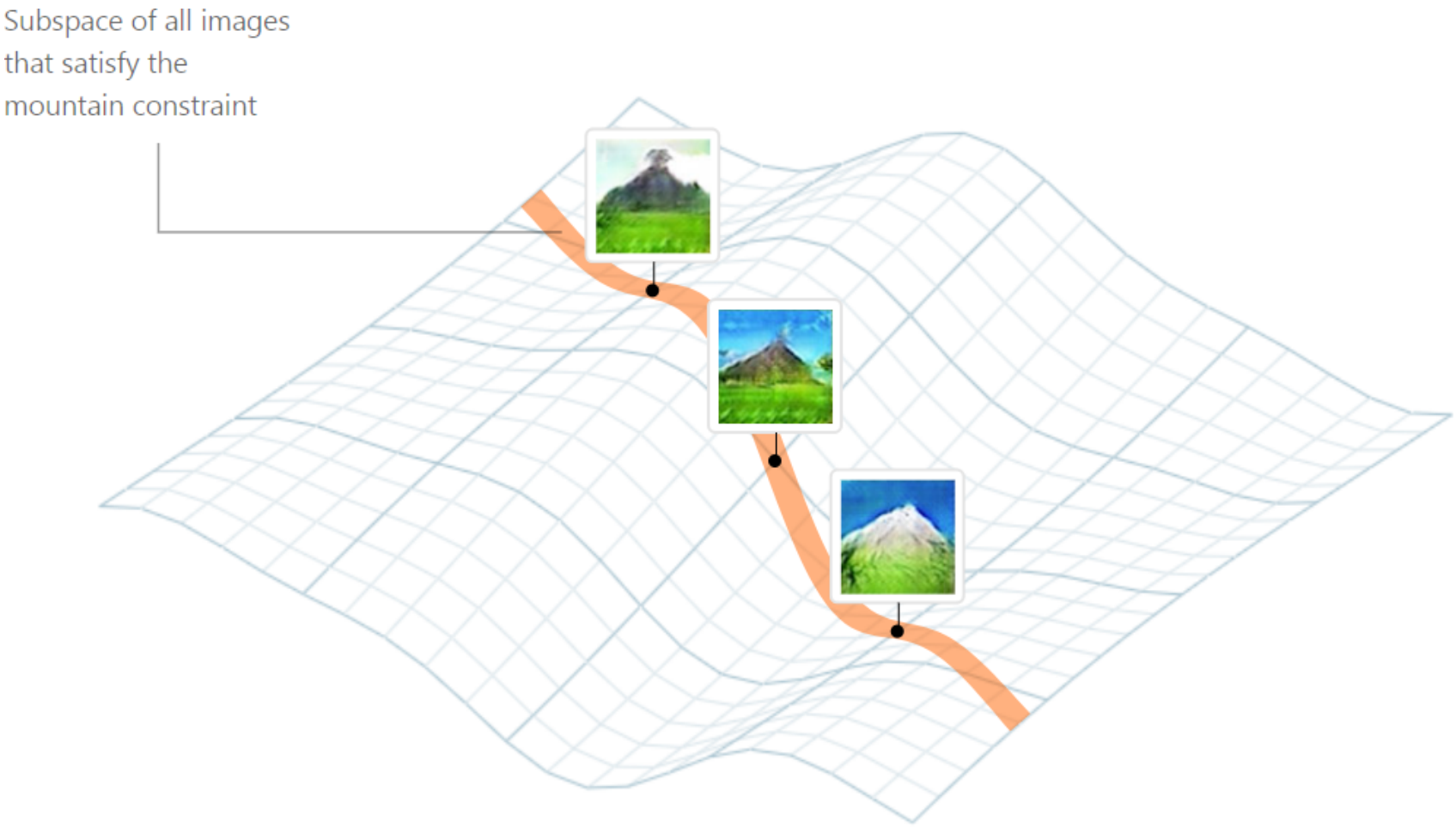

Favor simplicity

Let's play a game:

$$2, 4, 8, \ldots$$

Rules:

- **Task:** Discover the rule I used to produce the sequence
- Give me a number: I'll tell you if it's next in sequence or not
- **Once you're sure**, tell me the rule

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Occam's razor: Among competing hypotheses, select the one with the fewest assumptions.

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Occam's razor: Among competing hypotheses, select the one with the fewest assumptions.

Also: when feasible, add more data!

Suggested reading

Blog post on the datasaurus:
https://www.autodeskresearch.com/publications/samestats

TED talk on the idea behind imaging the black hole:
https://www.youtube.com/watch?v=BIvezCVcsYs

VLBI reconstruction dataset:
http://vlbiimaging.csail.mit.edu/

Paper on the black hole imaging technique:
https://arxiv.org/pdf/1512.01413.pdf

Tutorial video and slides on ML fairness:
https:
//nips.cc/Conferences/2017/Schedule?showEvent=8734

Distill post on t-SNE:
https://distill.pub/2016/misread-tsne/